1)Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

2. Create Sample Dataset

```
data = {
    'Month': ['Jan','Feb','Mar','Apr','May','Jun',
              'Jul','Aug','Sep','Oct','Nov','Dec'],
    'Sales': [12000,13500,14000,15000,16500,18000,
              19000,21000,22000,23500,25000,27000]
}

df = pd.DataFrame(data)

print(df.head()
```

3. Data Cleaning

```
# Check missing values
print(df.isnull().sum())

# Remove duplicates
df = df.drop_duplicates()

# Check data types
print(df.dtypes)

# Convert month into numeric index
df['Month_Num'] = range(1, len(df)+1)

print(df)
```

4. Exploratory Data Analysis

Summary Statistics

```
print(df.describe())
```

Total Sales

```
total_sales = df['Sales'].sum()
print("Total Sales:", total_sales)
```

```
Average Sales
avg_sales = df['Sales'].mean()
print("Average Sales:", avg_sales)

Highest Sales Month
highest = df.loc[df['Sales'].idxmax()]
print("Highest Sales Month:")
print(highest)

5) Sales Trend Visualization
plt.figure(figsize=(10,5))

plt.plot(
    df['Month'],
    df['Sales'],
    marker='o',
    linewidth=2
)

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)

plt.show()

6. Bar Chart
plt.figure(figsize=(10,5))

sns.barplot(
    x='Month',
    y='Sales',
    data=df
)

plt.title("Monthly Sales Comparison")

plt.show()

7. Prediction Model (Linear Regression)
X = df[['Month_Num']]
y = df['Sales']

model = LinearRegression()

model.fit(X, y)

predictions = model.predict(X)

print("R2 Score:", r2_score(y, predictions))
```

8. Forecast Future Sales

```
future_months = pd.DataFrame({
    'Month_Num':[13,14,15]
})

future_sales = model.predict(future_months)

forecast = pd.DataFrame({
    'Month':['Jan-2025','Feb-2025','Mar-2025'],
    'Predicted_Sales':future_sales
})

print(forecast)
```

9) Forecast Visualization

```
plt.figure(figsize=(10,5))

plt.plot(
    df['Month_Num'],
    df['Sales'],
    marker='o',
    label='Actual Sales'
)

future_x = [13,14,15]

plt.plot(
    future_x,
    future_sales,
    marker='o',
    linestyle='--',
    label='Forecast Sales'
)

plt.title("Sales Forecast")
plt.xlabel("Month Number")
plt.ylabel("Sales")
plt.legend()

plt.show()
```

10. Interactive Dashboard (Plotly)

```
import plotly.express as px

fig = px.line(
    df,
```

```
    x='Month',
    y='Sales',
    title='Sales Trend Dashboard',
    markers=True
)

fig.show()
```

11. KPI Dashboard Summary

```
print("========== SALES DASHBOARD ==========")

print("Total Sales:", df['Sales'].sum())

print("Average Sales:", round(df['Sales'].mean(),2))

print("Maximum Sales:", df['Sales'].max())

print("Minimum Sales:", df['Sales'].min())

growth = (
    (df['Sales'].iloc[-1] -
     df['Sales'].iloc[0])
    /
    df['Sales'].iloc[0]
) * 100

print("Growth Rate: {:.2f}%".format(growth))
```